Monitoring and Logging



What tools are there, what needs to be monitored?

Monitoring

Overview

- Command line tools
- Cloud provider provided tools
- Web dashboards
- Logging
- Custom dashboards: metrics

Command line tools

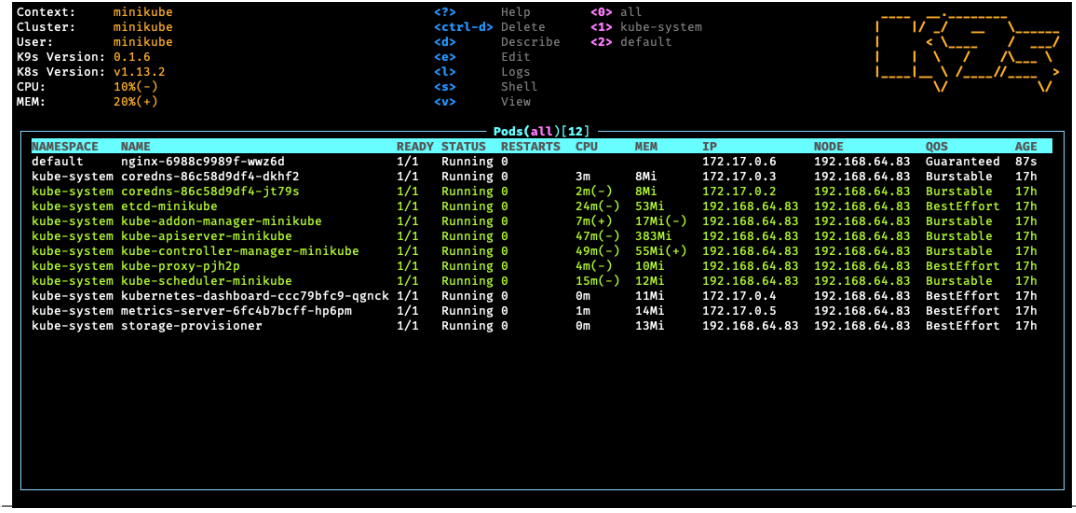
kubectl top

- Kubernetes provides metric collection (CPU/memory) for pods/nodes
- This has an API, that you can programmatically access
- But there is also command line access
  - `kubectl top pods`
  - `kubectl top nodes`

```
errge@brooks:~ $ kubectl top nodes
NAME                                CPU(cores)   CPU%   MEMORY(bytes)   MEMORY%
aks-default-35064155-vmss000006   101m         5%     760Mi           35%
aks-default-35064155-vmss000007   49m          2%     594Mi           27%
```

Command line tools

k9s



Command line tools

k8slens

- Similar to k9s, but graphical, good for non-ops people
- Runs on your computer, not a web dashboard
- https://k8slens.dev/

Cloud provider provided tools

GKE, Azure, AWS

- All cloud providers have their own logging, monitoring, dashboard
- Trade-off: portability vs integration
- It usually takes care of everything and have good default rules
- Easiest to set up

On-prem, self installed alternatives:

- Pixie: https://px.dev/
- Octant: https://octant.dev/

Logging

How to deal with `kubectl logs`

- Stored on the nodes, ephemeral, `/var/log/pods`
- To keep, need to archive it to a central server continuously
- Open source projects:
  - Fluentd + Elasticsearch
  - OpenTelemetry
  - Grafana Loki
- In the cloud, most probably you can enable this with one click

Custom dashboards: metrics

The Prometheus/Grafana stack

- Business vs ops monitoring
- Your apps have to provide metrics on a custom URL (instrumentation) (BPF may change this, e.g. Pixie)
- Prometheus collects and stores it, provides PromQL query language
- Grafana: draws dashboards from the PromQL expressions
- Alertmanager: email, PagerDuty, OpsGenie

PromQL playground: https://demo.promlabs.com/

Example: `sum(rate(demo_api_request_duration_seconds_count{path="/api/foo", method="GET"}[60m]))`

Thanks!